

The empirical recurrence for OEIS A251813, proved by transfer matrix

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Abstract

OEIS A251813 counts arrays of width 4 over $\{0, 1, 2\}$ in which every 3×3 window has constrained SUMS along its rows, its columns and its two diagonals. The entry carries an empirical recurrence of order 17 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a 3×3 window lies in 3 consecutive array rows, so the count is a walk count on $S = 6643$ states.

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1 The conjecture

OEIS A251813 is “Number of $(n+2) \times (2+2)$ 0..2 arrays with every row, column, diagonal or antidiagonal in each 3×3 subblock summing to a prime.”. It has offset 1 and begins

500, 538, 771, 1511, 3501, 9040, 24411, 66516, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = a(n-1) + 2*a(n-2) + 9*a(n-3) + 2*a(n-4) - 5*a(n-5) - 23*a(n-6) - 14*a(n-7) + 2*a(n-8) + 25*a(n-9) + 20*a(n-10) + 4*a(n-11) - 12*a(n-12) - 11*a(n-13) - 4*a(n-14) + 2*a(n-15) + 2*a(n-16) + a(n-17)$ for $n > 25$

As of the “Last modified” line on the live entry (Jul 23 2025 13:28:59, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 cells across, with entries in $\{0, 1, 2\}$. A WINDOW is a 3×3 block of consecutive cells lying inside the array. Every window must satisfy

the sum along	
each row	must be prime
each column	must be prime
the main diagonal	must be prime
the antidiagonal	must be prime

where the main diagonal of a window runs from its top left corner to its bottom right, and its antidiagonal from top right to bottom left.

3 The walk

Lemma 1. *A 3×3 window occupies 3 consecutive array rows, so every condition above is decided by 3 consecutive rows and by nothing else.*

Proof. By definition of a window. $\square$

So the state is the window of the last 2 array rows, a row not yet written carried as $\perp$, and a step appends a row and settles every 3×3 window that row completes. A window needing a row that is $\perp$ does not exist, which is the boundary rule; the walk starts at the all- $\perp$ state, so a walk of R steps is an array of R rows. With M the adjacency matrix and $\tau = \mathbf{1}$,

$$a(n) = \iota^\top M^j \tau, \quad S = 6643.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^{17} - \sum_i c_i t^{17-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+17} - \sum_i c_i M^{j+17-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = a(n-1) + 2a(n-2) + 9a(n-3) + 2a(n-4) - 5a(n-5) - 23a(n-6) - 14a(n-7) + 2a(n-8) + 25a(n-9) + 20a(n-10)$$

for every $n > 25$.

Proof. The vector $w = q(M) \tau$ was formed by 17 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 25 + 1$ onwards and the run continued until the working vector was identically zero. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order, in the orientation the entry's own name writes, and test a window as soon as its last cell is placed. No state, no transposition and no transfer matrix are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.